CLUE: Contrastive Learning with Uncertainty Estimation for Semi-supervised Medical Image Segmentation

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***Abstract*—Deep learning based medical image segmentation requires high quality pixel-level labeled data, which demands significant time and cost. Most existing semi-supervised learning methods exclude pseudo-labels with high uncertainty from training. This causes class imbalance and limits deep representation learning. In this study, we propose a semi-supervised medical image segmentation method based on contrastive learning that leverages uncertainty information. The proposed method achieved DSC and Jaccard scores of 89.48 and 81.50, with only 10% labeled data, surpassing existing methods.**

***Index Terms*—medical image segmentation, semi-supervised learning, contrastive learning, uncertainty estimation, medical assistant**

I. Introduction

Advances in artificial intelligence have enabled deep learning to deliver efficient and accessible medical services. Particularly medical image segmentation which provides pixel-wise classification, can help physicians to identify and localize the organs or tumors. However, such a task requires pixel-level labeling by experts, which incurs significant costs. Due to this reason, research focuses on semi-supervised learning that uses both unlabeled and limited labeled data to address the lack of labeled data in the medical field [1]. To effectively utilize unlabeled data in semi-supervised learning, methodologies such as pseudo-labeling and self-supervised learning have been proposed. Among them, FixMatch [2] employs a model to learn consistency between different augmentations applied to unlabeled images. As only weakly augmented images whose prediction score exceeds a threshold are used as pseudo-labels, it can cause class imbalance as some difficult yet important classes will be excluded.

On the other hand, contrastive learning is a type of self-supervised learning that maximizes the similarity between identical samples and minimizes it between different samples [4]. Previous studies performed contrastive learning on pseudo labels from medical images using the pixel-wise class information [5]. However, this approach may not consider the discriminative properties of features within the same class.

In this study, we propose a medical class-based contrastive learning method that utilizes uncertainty. We use image patches from pseudo-labels and calculate mean binary entropy of these patches as a measure of uncertainty. Then, only the features of patches with low uncertainty are used to calculate class distribution, and contrastive learning is applied to align features within each class to this distribution. Specifically, features with high uncertainty are given a higher weight when calculating the similarity with the distribution. This approach helps the model learn to reduce uncertainty even within the same class by reflecting the uncertainty among patch features, thus leading to fast convergence and performance improvement.

Experimental results show that the proposed method achieves a Dice Similarity Coefficient (DSC) of 89.48 and a Jaccard score of 81.50 when using 10% of the labels on the Automated Cardiac Diagnosis Challenge (ACDC) dataset, showing improved performance compared to existing models. The results demonstrate that effective and stable model learning by incorporating uncertainty is possible in environments with limited labeled data, leading to improved performance and offering valuable guidance for diagnosis and visualization

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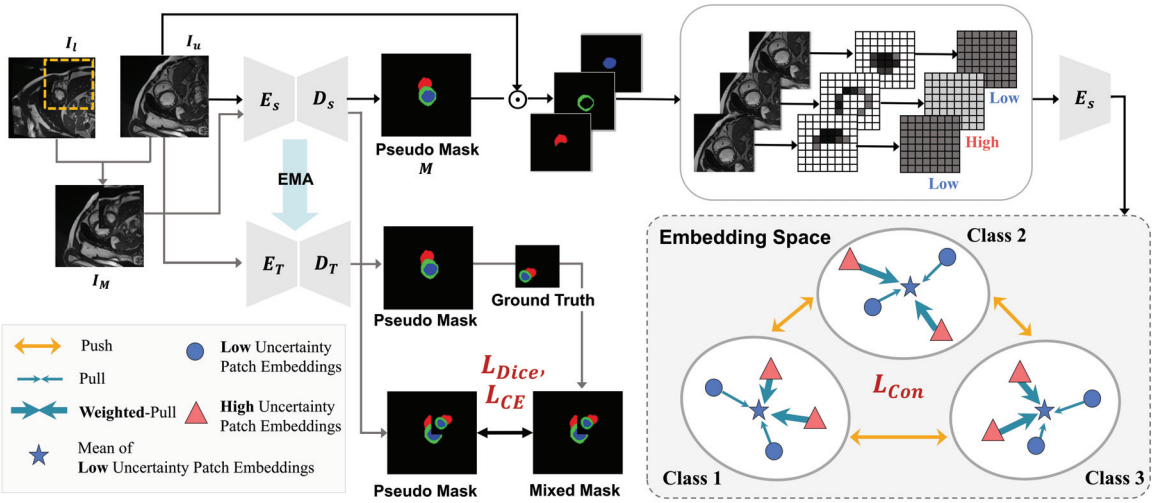

Fig. 1. Proposed contrastive learning framework for image segmentation.

in real-world medical applications.

II. Method

In this section, we explain the uncertainty-based contrastive learning method proposed in this paper. The proposed method simultaneously utilizes labeled images $I_L \in \mathbb{R}^{H\times W}$ and unlabeled images $I_U \in \mathbb{R}^{H\times W}$. Here, N_L and N_U denote the number of each type of image. Considering that we are dealing with semi-supervised learning with a small number of labeled images, the condition $N_L \ll N_U$ holds. The overall training process proposed in this paper is illustrated in Fig. 1.

A. Patch Sampling Based on Pseudo-Labels

An encoder and decoder network E, D are used to obtain mask information from the input image I_U. Here, the pairs E, D are divided into the teacher model E^T, D^T and the student model E^S, D^S. The pseudo-label mask $M_U \in \mathbb{R}^{K\times H\times W}$ is obtained using E^S and D^S. Here, K denotes the class, and H and W denote the height and width of the mask, respectively. A one-hot mask is generated by assigning 1 to the class mask with the highest probability value and 0 to the rest, thus representing only the region of a specific class K. $I_U^K = I_U \odot M_U^K \in \mathbb{R}^{H\times W}$ is the element-wise multiplication of the image and its corresponding one-hot mask, which is used for patch sampling. Subsequently, $M_U^K \in \mathbb{R}^{H\times W}$ is used to obtain contour information related to the region predicted for a specific class, and a bounding box is generated encompassing this region. Using the coordinates of the bounding box, I_U^K is cropped to extract the patch $P_U^K \in \mathbb{R}^{H^P\times W^P}$.

B. Extracting Uncertainty Within Patches

Next, binary cross-entropy is used for uncertainty estimation of patches. This metric evaluates the difference between two probability distributions in binary classification, with higher values indicating greater prediction uncertainty [6]. Consequently, The proposed method, calculates an uncertainty score for each class by using the expected value of binary cross-entropy for the corresponding patches. The equation to calculate s^K is as follows:

$$s^K = -\frac{1}{|H^P \times W^P|}\sum_i \left[p_i^K \log p_i^K + (1-p_i^K)\log(1-p_i^K)\right] \tag{1}$$

Here, p_i^K is the probability that the i-th pixel x_i within P_U^K belongs to class K. The s^K obtained from all patches of class K are divided into high uncertainty group S_{High} and low uncertainty group S_{Low} based on the median value. All patches of the background class, which does not directly affect the medical image segmentation, are classified into the low uncertainty group S_{Low}.

C. Contrastive Learning using Uncertainty Information

In the previous process, patches within images were classified based on class and uncertainty information. For contrastive learning, patch P_U^K is passed through the student model encoder E^S to generate a feature vector $f_U^K \in \mathbb{R}^v$ by reshaping the final feature map dimension:

$$f_U^K = E^S(P_U^K) \tag{2}$$

Here, v denotes the dimension of the feature vector. In the proposed contrastive learning method, the distance between all feature vectors f_U^K belonging to the same class and the distribution of the low uncertainty feature vector group is minimized, while the distance to all feature vectors belonging to different classes is maximized. The loss function for the proposed contrastive learning is calculated as follows. It assumes that the feature vectors of each class follow a normal

distribution [7]. Thus, $f_U^K \in \mathbb{R}^v$ follows a normal distribution with mean μ_{Low}^K and covariance σ_{Low}^K:

$$f_U^K \sim \text{Norm}(\mu_{\text{Low}}^K, \sigma_{\text{Low}}^K), \quad \text{where} \quad f_U^K \in S_{\text{Low}} \tag{3}$$

The similarity is calculated by scaling the dot product of the feature vectors by the temperature τ. Feature vectors belonging to the same class with respect to μ_{Low}^K are defined as positive, while those from different classes are defined as negative. The following equations represent the positive and negative similarities.

$$\text{sim}(f^1, f^2) = \frac{(f^1)^T f^2}{\tau} \tag{4}$$

$$\text{sim}^{K+} = \text{sim}(f_U^K, \mu_{\text{Low}}^K) \tag{5}$$

$$\text{sim}^{K-} = \text{sim}(f_U^K, \mu_{\text{Low}}^J), \quad K \neq J \tag{6}$$

When calculating positive similarity, the similarity with high certainty vectors is increased by adjusting the weight α for $f_U^K \in S_{\text{High}}$. Thus, the final loss function is as follows:

$$L_{\text{con}} = \log \Bigg[\exp\left(\text{sim}^{K+} + \frac{\lambda}{2\tau^2} (\text{sim}^{K+})^T \sigma_{\text{Low}}^K \text{sim}^{K+} \right) + \zeta \sum_{K \neq J} \exp\left(\text{sim}^{K-} + \frac{\lambda}{2\tau^2} (\text{sim}^{K-})^T \sigma_{\text{Low}}^J \text{sim}^{K-} \right) - \text{sim}^{K+} \alpha 1_{S_{\text{High}}}(f_U^K) \Bigg] \tag{7}$$

Here, ζ and λ are scaling parameters where ζ is the sum of all negative feature vectors for a specific class, and λ adjusts the effect of the feature vector distribution covariances σ_{Low}^K and σ_{Low}^J. By guiding features with high uncertainty to move more toward the class distribution, feature vectors are clearly separated along decision boundaries. The final loss function combines the Cross Entropy loss, commonly used for semantic segmentation, with the Dice loss, which accounts for segmentation accuracy. The equation is as follows:

$$L_{\text{total}} = w_1 \cdot L_{\text{con}} + w_2 \cdot L_{\text{ce}} + w_3 \cdot L_{\text{dice}} \tag{8}$$

Here, w_1, w_2, and w_3 are adjustable hyperparameters used to balance the contributions of each loss function.

III. Implementation

A. Dataset

To compare the proposed method with baselines, we used the ACDC dataset [8], which comprises short-axis cardiac MRI images with labels for three classes: left ventricle, right ventricle, and myocardium. The 100 cases in the dataset were divided into 70 for training, 20 for validation, and 10 for testing, in line with previous research [9]. The proportion of unlabeled data in the training data were set to 5% and 10%.

TABLE I
Comparison with semi-supervised segmentation baseline models

Method	Labeled	Metrics	
		DSC	Jaccard
DTC	3(5%)	56.90	45.67
URPC		55.87	44.64
MC-Net		62.85	52.29
SS-Net		65.83	55.38
BCP		87.59	78.67
Ours		**88.64**	**80.21**
DTC	7(10%)	84.50	73.92
URPC		84.29	72.41
MC-Net		83.10	77.04
SS-Net		86.44	77.67
BCP		88.84	80.62
Ours		**89.48**	**81.50**

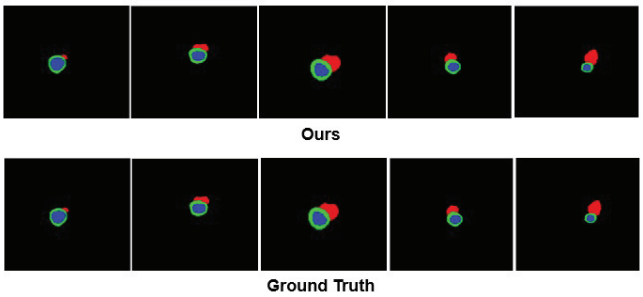

Fig. 2. The visualization of the predicted segmentation map using the proposed method.

B. Training Setup and Evaluation Metrics

In the experimental process, the size of 2D images was resized to 256×256, and the patch size was fixed at 128×128. The proposed method is based on the U-Net structure [10], with a batch size of 24 during training. SGD was used as the optimizer for model pre-training, and the learning rate was set to 0.01. The weights w_1, w_2, and w_3 used in the loss function were set to 1, 1, and 0.5, respectively. The baseline models used to verify the effectiveness of the proposed method include DTC [11], URPC [12], MC-Net [13], SS-Net [14], and BCP [9]. The baseline model BCP in this paper predicts labels for images generated by mixing labeled and unlabeled data to reduce the semantic distribution gap between the two types of data. The evaluation metrics used are DSC and Jaccard score. The DSC score is the intersection divided by the sum of the individual regions, and the Jaccard score is the intersection divided by the union of the two regions.

C. Experimental Results and Analysis

Table 1 shows that our method outperformed the baselines, achieving DSC and Jaccard scores of 88.64 and 80.21 with 3 labels, and 89.48 and 81.50 with 7 labels. These results confirm that guiding high-uncertainty features closer to low-uncertainty distributions improves medical image segmentation performance.

Fig. 2 illustrates the segmentation results of the proposed model for three predicted labels in the ACDC dataset. The

TABLE II
SCORE COMPARISON BASED ON UNCERTAINTY CONTRASTIVE LEARNING SIMILARITY ENHANCEMENT WEIGHT α

Labeled	α	1.5	2	3
3(5%)	DCS	88.36	**88.64**	88.26
	Jaccard	79.77	**80.21**	79.22
7(10%)	DCS	89.35	**89.48**	89.22
	Jaccard	81.31	**81.50**	81.26

TABLE III
SCORE COMPARISON BASED ON UNCERTAINTY CONTRASTIVE LOSS FUNCTION WEIGHT

w_1	0.3	0.5	0.7	0.9
DCS	86.82	**88.37**	88.34	88.21
Jaccard	77.39	**79.83**	79.69	79.51

method closely matches the Ground Truth for each label, notably achieving smooth segmentation in the typically challenging boundary regions between classes.

The results of segmentation performance comparison based on the weight α are presented in Table 2. The experiment results indicate that the best performance was achieved when the number of labels used for training was 3 and 7, and α was set to 2. Specifically, when the number of labels was 3, the DSC was 88.64 and the Jaccard score was 80.21, while with 7 labels, the DSC was 89.48 and the Jaccard score was 81.50. This suggests that adjusting only the weight α for high-uncertainty patch feature vectors within the same class results in a significant performance gap. Particularly, when α was set to 3, the performance was not as good as when it was set to 1.5. This indicates that setting α too high might lead to an overemphasis on the class information, beyond merely adjusting the uncertainty of patch feature vectors.

The results of segmentation performance comparison based on the weight w_1 of the uncertainty contrastive loss function are shown in Table 3. In the environment with 7 labels, the highest performance was observed when w_1 was 0.5, with a DSC of 88.37 and a Jaccard score of 79.83. It was confirmed that merely adjusting the weight within the proposed loss function improved the scores by 2.05 and 2.44, respectively, compared to when w_1 was 0.3. As no significant performance difference was observed with higher weights, it was verified that setting w_1 to an appropriate size, considering the influence of Cross Entropy and Dice loss functions, is essential.

IV. CONCLUSION

In this paper, we proposed a semi-supervised learning method based on contrastive learning using uncertainty for medical image segmentation. With utilizing uncertainty information, the distance between high-uncertainty patches and the distribution of corresponding class was minimized, resulting in better separation for given classes and effective representation learning. Experimental results showed that with 10% labeled data, the DSC scores reached 89.48 and 88.64, and the Jaccard scores reached 81.50 and 80.21, respectively. This demonstrate that incorporating feature vector uncertainty during contrastive learning allows for efficient and stable model training in data-scarce environments. We anticipate that this approach could be helpful for real-world applications, such as diagnosis and ROI (region of interest) visualization, where labeled data are limited and effective use of unlabeled images is crucial.

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